Supplemental Material: Observable relaxation and perturbation detection near a fermionic Liouvillian exceptional point

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I. CONDITIONAL GENERATOR AND PHYSICAL PREPARATION

For $n \geq 2$ modes, $1 \leq d \leq n$, and a number-conserving quadratic Hamiltonian $H = \sum_{ij} h_{ij} c_i^\dagger c_j$, $h = h^\dagger$, the retained state satisfies $\dot{\widetilde{\rho}} = \mathcal{L}\widetilde{\rho}$, where

$$\mathcal{L}\rho = -\mathrm{i}[H,\rho] + \gamma \sum_j c_j \rho c_j^\dagger - n\gamma\rho + z\gamma \sum_{|I|=d} c_I^\dagger \rho c_I. \tag{S1}$$

Here $\gamma > 0$ and $c_I^\dagger = c_{i_1}^\dagger \cdots c_{i_d}^\dagger$ for $i_1 < \cdots < i_d$. All subsets have equal weight. The pump is recycling only, as in Xu and Yi [1]; adding pump anticommutators changes the generator.

For $z \geq 0$, the recycling terms are completely positive (CP), and the no-recycling evolution is a scalar attenuation times a unitary channel. Their Dyson expansion is therefore CP. Counting holes gives

$$\mathcal{L}^\dagger I = \gamma \left[-(n - \hat{N}) + z \binom{n-\hat{N}}{d} \right]. \tag{S2}$$

For h holes, trace nonincrease requires $z\binom{h}{d} \leq h$. For $h \geq d$, the ratio $\binom{h}{d}/h = \binom{h-1}{d-1}/d$ is largest at $h = n$; for $h < d$, the condition is automatic. Thus the exact physical range and conditional normalization are

$$0 \leq z \leq \frac{n}{\binom{n}{d}},$$

$$P_{\mathrm{surv}}(t) = \mathrm{Tr}\,\widetilde{\rho}(t), \qquad \rho(t) = \frac{\widetilde{\rho}(t)}{P_{\mathrm{surv}}(t)}. \tag{S3}$$

The positive jump effect

$$R = \gamma \left[n - \hat{N} - z \binom{n-\hat{N}}{d} \right] \tag{S4}$$

supplies the discarded branch of a monitored completion. For the endpoint channel studied below, every rejected system is replaced by one orthogonal erasure flag. A retained input ancilla is not discarded by this completion.

Use energy eigenmodes to define $n_j = c_j^\dagger c_j$, $\hat{N} = \sum_j n_j$, $D = n_1 - n_2$, and $\Pi = (-1)^{\hat{N}}$. The common initial state is

$$\rho_0 = \rho_p(1 + \epsilon D), \qquad \rho_p = p^{\hat{N}}(1-p)^{n-\hat{N}}. \tag{S5}$$

For $0 < p < 1$ and $0 < |\epsilon| < 1$, ρ_p has full support and the factors $1 + \epsilon D$ are strictly positive. Exchange symmetry gives $\mathrm{Tr}(\rho_p D) = 0$, so ρ_0 is normalized. It commutes with Π and contains both particle-number parities. Its initial observables are $\langle \hat{N} \rangle_0 = np$ and $\langle D \rangle_0 = 2\epsilon p(1-p)$. Equation (S5) is exactly the probability reweighting described in the main text: the 10 and 01 weights are multiplied by $1+\epsilon$ and $1-\epsilon$.

II. EXACT OBSERVABLE PROJECTIONS AND VISIBLE RATES

The two density signals reduce exactly to two population blocks with a common survival normalization. To derive this reduction, identify the diagonal operator space with n Boolean spins, one for the occupation of each energy eigenmode. Define $J_\pm = \sum_j \sigma_j^\pm$, where $\sigma_j^\pm$ raise or lower the jth Boolean occupation spin. In this space, the recycling generator is $G_{n,d} = J_- + zJ_+^d/d!$. In the basis of unnormalized particle-number projectors P_k, its maximal-spin matrix $N_{n,d}(z)$ has entries

$$N_{k-1,k} = n - k + 1, \qquad N_{k+d,k} = z\binom{k+d}{d}. \tag{S6}$$

These are coefficients per configuration. For any effective size $M \geq 0$, the aggregate-population matrix is

$$B_{M,d} = W_M N_{M,d} W_M^{-1}, \qquad (W_M)_{kk} = \binom{M}{k},$$
$$[B_{M,d}(z)]_{k-1,k} = k, \qquad [B_{M,d}(z)]_{k+d,k} = z\binom{M-k}{d}. \tag{S7}$$

All other entries are zero, and out-of-range transitions are absent. The binomial factors are essential when calculating probabilities or expectation values. At zero pump, the nonzero lowering chain obeys $B_{n,d}(0)^{n+1} = 0$ and $B_{n,d}(0)^n \neq 0$: it forms a single size-$(n+1)$ Jordan block, placing the population EP at $-n\gamma$.

For $D = n_1 - n_2$, let $m = n - 2$ and $Q_k = DP_{k+1}$. The two marked Boolean spins form the singlet $|10\rangle_B - |01\rangle_B$, while the other m spins are symmetric. Thus $Q_k = J_+^k Q_0/k!$ and

$$J_- Q_k = (m-k+1)Q_{k-1}, \qquad J_+ Q_k = (k+1)Q_{k+1},$$
$$\mathrm{Tr}\,Q_k = \mathrm{Tr}(\hat{N}Q_k) = 0, \qquad \mathrm{Tr}(DQ_k) = 2\binom{m}{k}. \tag{S8}$$

The span of the Q_k is one spin-$m/2$ copy. Since it is invariant under both J_+ and its transpose J_-, its orthogonal complement in the Boolean space is also invariant.

As $D = \sum_k Q_k$, other spin copies do not contribute to this observable.

For an arbitrary initial density matrix define the aggregate projections

$$u_k = \mathrm{Tr}(P_k \rho_0), \qquad v_k = \mathrm{Tr}(D P_{k+1} \rho_0). \tag{S9}$$

Writing $E_M(t) = e^{\gamma t B_{M,d}(z)}$, the exact signals are

$$\langle \hat{N} \rangle_t = \frac{\boldsymbol{k}^T E_n(t) u}{\mathbf{1}^T E_n(t) u},$$
$$\langle D \rangle_t = \frac{\mathbf{1}^T E_m(t) v}{\mathbf{1}^T E_n(t) u}, \qquad P_{\mathrm{surv}} = e^{-n\gamma t} \mathbf{1}^T E_n(t) u, \tag{S10}$$

where $\boldsymbol{k} = (0, 1, \ldots, n)^T$. Off-diagonal energy-basis operators remain in complementary invariant blocks and have zero trace against these diagonal observables.

For $\rho_0 = \rho_p(1 + \epsilon D)$, the two projections become

$$u_k = \binom{n}{k} p^k (1-p)^{n-k},$$
$$v_k = 2\epsilon \binom{m}{k} p^{k+1} (1-p)^{n-k-1}. \tag{S11}$$

Equivalently, let $F_{M,k} = \binom{M}{k} p^k (1-p)^{M-k}$ and $R_M(t) = E_M(t) F_M$. The signals plotted in the main text are

$$\frac{\langle \hat{N} \rangle_t}{n} = \frac{\sum_{k=0}^n (k/n) R_{n,k}(t)}{\sum_{k=0}^n R_{n,k}(t)},$$
$$\langle D \rangle_t = 2\epsilon p(1-p) \frac{\sum_{k=0}^{n-2} R_{n-2,k}(t)}{\sum_{k=0}^n R_{n,k}(t)},$$
$$P_{\mathrm{surv}}(t) = e^{-n\gamma t} \sum_{k=0}^n R_{n,k}(t). \tag{S12}$$

Changing ϵ leaves the entire number and survival curves unchanged and scales D linearly. The positivity of u, and positivity of $\mathrm{sgn}(\epsilon) v$, guarantee their Perron overlaps when the corresponding matrices are irreducible. At $p = 0.75$, $\epsilon = 0.6$, $\langle D \rangle_0 = 0.225$ and $\langle \hat{N} \rangle_0 = 0.75 n$.

For $z > 0$, put $q = z^{1/(d+1)}$. The diagonal matrix $\mathsf{D}_{kk} = q^k$ gives $\mathsf{D}^{-1} B_{M,d}(z) \mathsf{D} = q B_{M,d}(1)$. Let $\kappa_{M,d}$ be the Perron root of $B_{M,d}(1)$, with $\kappa_{M,d} = 0$ for $M < d$, and define

$$\delta_{n,d} = \kappa_{n,d} - \max_{\nu \in \mathrm{spec}\, B_{n,d}(1),\, \nu \neq \kappa_{n,d}} \Re\nu. \tag{S13}$$

The leading full eigenvalue is $\lambda_* = -n\gamma + \gamma q \kappa_{n,d}$. For $n - 2 \geq d$, the imbalance has a nonzero leading overlap and

$$\langle D \rangle_t \sim A_D e^{-\Gamma_D t}, \quad \Gamma_D = \gamma q (\kappa_{n,d} - \kappa_{n-2,d}),$$
$$\langle \hat{N} \rangle_t / n - \nu_* \sim A_N e^{-\Gamma_N t}, \quad \Gamma_N = \gamma q \delta_{n,d}. \tag{S14}$$

The second line applies only when a real subleading root determining $\delta_{n,d}$ has nonzero normalized particle-number overlap. The more general overlap criterion follows next.

The particle-number rate is fixed by the modes that survive normalization and have nonzero overlap with the preparation. Let r_j, l_j be right and left eigenvectors of $B_{n,d}(z)$ at a simple root μ_j, with $l_j^\dagger r_j = 1$. Set $\mu_* = q\kappa_{n,d}$ and $\nu_* = (\boldsymbol{k}/n)^T r_* / (\mathbf{1}^T r_*)$. Then

$$C_j = (\boldsymbol{k}/n - \nu_* \mathbf{1})^T r_j \, l_j^\dagger u. \tag{S15}$$

The conditional rate uses the largest $\Re\mu_j$ for which $C_j \neq 0$, with $j \neq *$. The denominator's leading amplitude $(\mathbf{1}^T r_*)(l_*^\dagger u)$ is strictly positive for any physical nonzero $u \geq 0$. A nonzero numerator is a separate requirement. For real simple first and second roots,

$$A_N = \frac{C_1}{(\mathbf{1}^T r_*)(l_*^\dagger u)}, \qquad A_D = \frac{(\mathbf{1}^T r_{m,*})(l_{m,*}^\dagger v)}{(\mathbf{1}^T r_*)(l_*^\dagger u)}. \tag{S16}$$

The reported theoretical curves use these projections, not fitted amplitudes. Only the relevant simple roots are projected: inverting a complete eigenvector matrix would be invalid in the presence of a residual zero Jordan block.

For $O_\eta = \hat{N}/n + \eta D/n$, the two-channel tail is

$$\langle O_\eta \rangle_t - \nu_* \simeq A_N e^{-\Gamma_N t} + \frac{\eta A_D}{n} e^{-\Gamma_D t}. \tag{S17}$$

Equating the two asymptotic magnitudes gives

$$t_\times \simeq \frac{\log[n|A_N|/(|\eta A_D|)]}{\Gamma_N - \Gamma_D}. \tag{S18}$$

For the main-text dynamical example, $A_N = .726392$ and $A_D = .235641$. With $\eta = .1$, this gives $\gamma t_\times \simeq 10.54$. The formula assumes separated real exponents and nonzero amplitudes; it describes their competition rather than a new selection mechanism.

If $m < d$, the imbalance block contains only lowering transitions. The positive initial state has a nonzero highest-weight component, so its numerator has degree m in time. Its long-time form is $A_D t^m e^{-\gamma q \kappa_{n,d} t}$, not a pure exponential. Complex visible roots likewise require oscillatory factors. These cases do not invalidate the exact expressions (S10).

Observable-sector selection is a pre-existing mechanism [2]. The present specialization provides exact fermionic channels and their conditional normalization. Fermion-parity evenness means $\Pi\rho\Pi = \rho$, not exclusive support on even particle numbers. One-time even-state/even-observable signals have no odd component. A two-time fermionic correlator can have an odd intermediate insertion, and is outside that exclusion.

III. COMPLETE FERMIONIC BLOCKS AND GAPS

Diagonalize the number-conserving quadratic Hamiltonian. Both the uniform loss channels and all uniform d-creation channels are invariant under this rotation; the

latter statement uses unitarity of the exterior-power representation. Let A, B be disjoint ket-only and bra-only occupied sets and C a common occupied subset of the remaining $m = n - |A| - |B|$ modes. The operators $|A \cup C\rangle\langle B \cup C|$ have Hamiltonian shift $-\mathrm{i}(E_A - E_B)$. Multiplication by

$$g_{A,B}(C) = (-1)^{\sum_{c\in C} |\{f\in A\cup B : f<c\}|} \tag{S19}$$

removes the fermionic signs from every recycling edge. The block is

$$[-n\gamma - \mathrm{i}(E_A - E_B)]\mathbb{1} + \gamma G_{m,d}. \tag{S20}$$

The spin-$s = (m-2j)/2$ component of $G_{m,d}$ is similar to $N_{m-2j,d}$ and hence to $B_{m-2j,d}$. Its multiplicity is $\eta_{m,j} = \binom{m}{j} - \binom{m}{j-1}$. The full multiset spectrum is

$$\biguplus_{\substack{A\cap B=\varnothing \\ 0\le j\le \lfloor m/2\rfloor}} [-n\gamma - \mathrm{i}(E_A - E_B) + \gamma \operatorname{spec} B_{m-2j,d}(z)]^{\eta_{m,j}}. \tag{S21}$$

The Hamiltonian shifts do not affect real-part gaps. Quadratic-fermion Jordan theory [3] does not directly diagonalize the $d > 1$ recycling term; the uniform-channel representation is used here.

For $m \ge d$, the directed Boolean graph of $G_{m,d}$ is irreducible. Any set can lose all its particles; from the empty set one can construct any set by additions of d particles and removals of the surplus. Its Perron vector is unique and permutation invariant, hence belongs to maximal spin. The subspace of subsets not containing an added mode gives $G_{m,d}$ as a principal submatrix of $G_{m+1,d}$. Perron monotonicity then orders the block edges. Parity-even operators require $|A| + |B|$ even, and the largest competing edge has effective size $n-2$ rather than $n-1$. Competition with the maximal block's own subleading roots gives the exact real-part gaps

$$\begin{aligned} \Delta_{\mathrm{all}} &= \gamma q \min\{\delta_{n,d}, \kappa_{n,d} - \kappa_{n-1,d}\}, \\ \Delta_{\mathrm{even}} &= \gamma q \min\{\delta_{n,d}, \kappa_{n,d} - \kappa_{n-2,d}\}. \end{aligned} \tag{S22}$$

Neither minimum can generally be removed. For the main-text example $(n, d, z) = (12, 3, 1/144)$,

$$(\Delta_{\mathrm{all}}, \Gamma_D, \Gamma_N)/\gamma = (0.571963,\ 1.117616,\ 1.678630), \tag{S23}$$

with $\Delta_{\mathrm{even}} = \Gamma_D$ and nonzero particle-number overlap.

For $n = d = 4$, $N_{4,4}(1)^5 = 24\mathbb{1}$. Its subleading fifth roots have real part $24^{1/5}\cos(2\pi/5) > 0$, whereas $\kappa_{3,4} = \kappa_{2,4} = 0$. Thus the internal complex roots control both gaps, and the D numerator has a quadratic time prefactor. This physical counterexample ($z = 1 \le 4$) rules out a universal identification of Γ_D with Δ_{even}.

IV. POLYNOMIAL TRUNCATION AND ESTABLISHED SPECTRAL STRUCTURE

The characteristic polynomial resolves the full finite-size root pattern and identifies the range of the linear pump truncation in Ref. [1]. Define $p_\ell(\mu, z)$ as the determinant of the first ℓ rows and columns of $\mu\mathbb{1} - N_{n,d}(z)$, with the full n fixed. Then $p_0 = 1$, $p_\ell = \mu^\ell$ for $1 \le \ell \le d$, and

$$\begin{aligned} p_\ell &= \mu p_{\ell-1} - z w_{\ell-d-1} p_{\ell-d-1}, \\ w_i &= \binom{i+d}{d} \frac{(n-i)!}{(n-i-d)!}. \end{aligned} \tag{S24}$$

A nontrivial cycle in a determinant permutation occupies a consecutive interval of length $d+1$. The disjoint-interval expansion gives

$$p_{n+1} = \sum_{j=0}^{\lfloor (n+1)/(d+1)\rfloor} (-z)^j W_j \mu^{n+1-(d+1)j}, \qquad W_j > 0, \tag{S25}$$

where $W_0 = 1$. Coefficients of this type and all-orders leading splitting polynomials are established in structured-Jordan perturbation theory [4]; higher-spin perturbations and eigenvalue rings also have direct precedents [5].

With $q = z^{1/(d+1)}$, the diagonal similarity $\mathsf{D}_{kk} = q^k$ gives $\mathsf{D}^{-1} N(z) \mathsf{D} = qN(1)$. Thus every nonzero eigenvalue has exact fractional scaling. McMillen's theorem [6] gives distinct positive radii ρ_a and simple nonzero roots $\rho_a e^{2\pi \mathrm{i} j/(d+1)}$. The zero algebraic multiplicity is $r = (n+1) \bmod (d+1)$. The matrix is cyclic, since its transpose is unreduced upper Hessenberg; for $r > 0$, the residual zero generalized eigenspace is one size-r Jordan block.

For $n = 7, d = 2$, one obtains

$$p_8 = \mu^8 - 504 z \mu^5 + 27504 z^2 \mu^2. \tag{S26}$$

There are two cubic triplets and one size-two zero Jordan block (Fig. S1). The linear truncation gives one triplet and five zero roots. At fixed nonzero μ, the discarded term is $O(z^2)$; at $\mu = qx$, every term in Eq. (S25) is proportional to q^{n+1}. The truncation is nonuniform on the splitting scale. Higher powers first occur at $n + 1 \ge 2(d+1)$.

Eq. (54) of Xu and Yi [1] uses an approximate-equality sign (arXiv v2 numbering; Eq. (49) in v1). This observation limits extraction of the full root pattern from that truncation; it does not overturn their exponent. Their Fig. 2 uses $n = 4$, where the truncation is complete for $d = 2, 3, 4$, while Appendix C separately solves $d = 1$. Those tests are not contradicted. Publication metadata were verified; the equation-by-equation check uses arXiv v2 because the publisher PDF was not accessible.

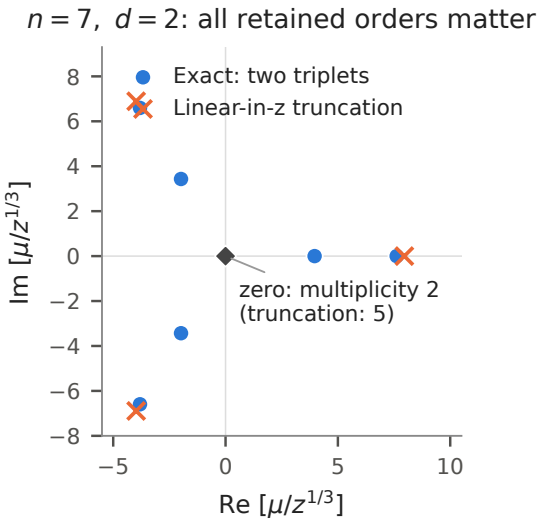


FIG. S1. Exact scaled roots for $n = 7, d = 2$ and the linear-in-z truncation. The discrepancy is independent of $z > 0$ after dividing out $z^{1/3}$. The exact residual zero multiplicity is two.

V. SEMICLASSICAL EXPANSION AND ITS LIMITS

At fixed d, define

$$\alpha_d = \frac{2d}{d+1}, \qquad \beta_d = \frac{d-1}{d+1}, \quad \Omega_d = \sqrt{\frac{d+1}{2}},$$
$$a_d = \left[\frac{2^{-(d-1)}}{(d-1)!}\right]^{1/(d+1)}, \qquad C_d = \frac{a_d}{\alpha_d}. \tag{S27}$$

Let $s = n/2$ and use the orthonormal spin representation $\mathsf{N}_s = J_- + zJ_+^d/d!$. With $S = e^{\eta J_z}$ and $e^{(d+1)\eta} = zs^{d-1}/(d-1)!$,

$$S^{-1}\mathsf{N}_s S = e^{\eta} T_s, \qquad T_s = J_- + \frac{J_+^d}{ds^{d-1}}. \tag{S28}$$

Around the $+x$ direction, set $J_x = s - Q$, $Q = b^\dagger b = (X^2 + P^2 - 1)/2$, $[X, P] = \mathrm{i}$, and $J_y = \sqrt{s}X + O(s^{-1/2})$. Keeping noncommutative operator ordering gives

$$T_s = s(1 + 1/d) + H_0 + s^{-1/2}V_3 + O(s^{-1}),$$
$$H_0 = 1 - P^2 - \frac{d+1}{2}X^2,$$
$$V_3 = -\mathrm{i}(d-1)\left[\frac{\{X, Q\}}{2} + \frac{d-2}{6}X^3\right]. \tag{S29}$$

The constant $+1$ comes from $-2Q = 1 - X^2 - P^2$. The leading odd operator V_3 has zero diagonal expectation in oscillator eigenstates. Its second-order correction enters at $O(s^{-1})$, together with the next even operator. Since $e^{\eta} = qa_d n^{\beta_d}$, the oscillator levels $1 - (2\ell + 1)\Omega_d$ give, for fixed low index ℓ, the formal edge ladder

$$\mu_\ell = q\left\{C_d n^{\alpha_d} + a_d[1 - (2\ell + 1)\Omega_d]n^{\beta_d} + O(n^{\beta_d - 1})\right\}. \tag{S30}$$

Identifying its first two levels with the leading positive radial roots predicts

$$\delta_{n,d} \sim 2a_d\Omega_d n^{\beta_d},$$
$$\kappa_{n,d} - \kappa_{n-r,d} \sim r a_d n^{\beta_d}, \qquad r \text{ fixed}. \tag{S31}$$

For fixed $d > 1$, inserting these in Eqs. (S14) and (S22), with the stated observable overlap, gives

$$\Delta_{\mathrm{all}} : \Gamma_D : \Gamma_N \longrightarrow 1 : 2 : \sqrt{2(d+1)}, \qquad \Delta_{\mathrm{even}} \sim \Gamma_D. \tag{S32}$$

For $d = 1$, $\mu_\ell = q(n - 2\ell)$ exactly, and the ratio is $1 : 2 : 2$ whenever the particle-number transient is nonzero.

Independently, the discrete ansatz $v_k \sim e^{nS(k/n)}$ at $z = \zeta n^{1-d}$ gives

$$\frac{\mu}{n} = (1 - x)e^p + \frac{\zeta x^d}{d!}e^{-dp}, \qquad p = S'(x). \tag{S33}$$

Minimizing over real p and maximizing the envelope over x yields $x_* = 1/2$ and $\mu/n \sim C_d\zeta^{1/(d+1)}$, where

$$C_d = (d+1)d^{-d/(d+1)}(d!)^{-1/(d+1)}4^{-d/(d+1)}. \tag{S34}$$

The saddle location is not an assertion that the conditional particle filling is one half: the coefficient vector and population vector differ by Eq. (S7).

These are formal fixed-d, fixed-low-excitation asymptotics. The expansion is not a global operator-norm estimate. McMillen's exact rays exclude off-axis edge competitors, whose real-part separation is of order n^{α_d} rather than n^{β_d}. They do not prove that the first two positive radial roots coincide with the local oscillator levels, or a uniform remainder bound. The latter identification is supported by Fig. S2 and is explicitly retained as an assumption in deriving the large-n gap hierarchy.

For $d = 2$, $N_{n,2}(2)^T$ is the $a = 0$ matrix of Shapiro and Tater [7]. Their radius $\frac{3}{4}n^{4/3}[1 + o(1)]$ gives $C_2 = 3/2^{7/3}$ by exact rescaling, so this leading coefficient is established prior art. The general fixed-d first correction and local spacing provide the additional ingredients for the complete fermionic-sector hierarchy under the assumptions stated above.

To remain within the trace-nonincreasing domain as n grows, take $z = \zeta n^{1-d}$ with fixed $0 < \zeta < d!$. Equations (S14) and (S30) predict

$$\Gamma_D \sim 2\gamma a_d\zeta^{1/(d+1)},$$
$$\Gamma_N \sim 2\gamma a_d\Omega_d\zeta^{1/(d+1)},$$
$$\frac{\lambda_*}{\gamma n} = -1 + C_d\zeta^{1/(d+1)} + O(n^{-1}). \tag{S35}$$

For $d > 1$, $C_d(d!)^{1/(d+1)} < 1$. Thus the asymptotic survival exponent remains extensively negative although the conditional rates remain finite. This is a statement about long-time survival and does not impose an exponential preparation cost for detecting the pump.

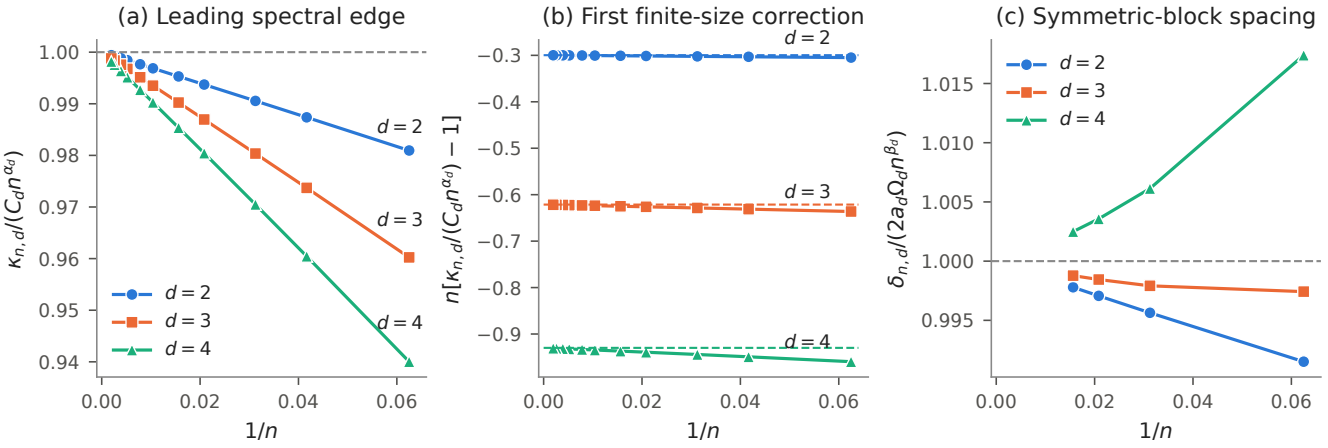


FIG. S2. Independent tests of the leading edge, first correction, and maximal-block spacing for $d = 2, 3, 4$. (a) Perron root divided by $C_d n^{\alpha_d}$. (b) Scaled residual tending to $\alpha_d(1 - \Omega_d)$. (c) Internal real-part gap divided by $2a_d\Omega_d n^{\beta_d}$. No coefficient is fitted. The $z = 1$ values can be rescaled into the trace-nonincreasing domain. Positive iteration extends through $n = 512$; dense arbitrary-precision spacings extend through $n = 64$.

VI. ENDPOINT STATISTICS AND SHORT-TIME DETECTION

Endpoint statistics determine the preparation cost of distinguishing two known pump strengths. Set $a = z\binom{n}{d}/n \in [0, 1]$ and $\tau = \gamma t$. The retained population generator is $L_a = B_{n,d}(an/\binom{n}{d}) - nI$. Appending an absorbing failure state gives

$$Q_a(f, k) = n - k - \frac{an}{\binom{n}{d}}\binom{n-k}{d}, \qquad Q_a(f, f) = 0. \tag{S36}$$

All other entries in the failure column vanish. Column sums are zero, off-diagonal entries are nonnegative, and every retained state has total exit rate n. The endpoint outcomes are $y \in \{0, \ldots, n, f\}$. The initial state and time are common to both hypotheses, and failed attempts count toward the preparation budget. The parameter a is distinct from the semiclassical coefficient a_d and the scaling parameter ζ.

A. Initial-state optimization for number readout

For any density matrix, $u_k = \mathrm{Tr}(P_k\rho_0)$ satisfies the closed population equation

$$\frac{du_j}{d\tau} = (j+1)u_{j+1} + \frac{an}{\binom{n}{d}}\binom{n-j+d}{d}u_{j-d} - nu_j, \tag{S37}$$

with out-of-range terms zero. This follows from closure of $\mathrm{span}\{P_k\}$ under $\mathcal{L}^\dagger$. It therefore holds beyond diagonal initial states, including physical parity-even states with particle-number coherences.

Let $T_i^{(k)}$ denote the endpoint distribution for initial particle number k and hypothesis a_i. Define

$$\mathrm{BC}(p, q) = \sum_y \sqrt{p_y q_y},$$
$$H^2(p, q) = 1 - \mathrm{BC}(p, q), \qquad \xi = -\log \mathrm{BC}\,. \tag{S38}$$

The endpoint distributions from a common mixture are $p_i^u = \sum_k u_k T_i^{(k)}$. Joint concavity gives

$$\begin{aligned}\mathrm{BC}(p_0^u, p_1^u) &\geq \sum_k u_k \,\mathrm{BC}(T_0^{(k)}, T_1^{(k)}) \\ &\geq \min_k \mathrm{BC}(T_0^{(k)}, T_1^{(k)}).\end{aligned} \tag{S39}$$

Thus some fixed-k input optimizes H^2 or ξ at each time. Enumeration of all k remains sufficient when optimizing time. This does not establish vacuum optimality for every parameter pair. It assumes no additional mean-number, full-rank, or state-preparation-cost constraint. A fixed odd-k density matrix is parity even as an operator. Conditioning on success alone would invalidate the mixture argument.

B. Vacuum protocol and its one-pump coefficient

For $\tau = c/n$, uniformization gives

$$e^{\tau Q_a} = e^{-c}\sum_{m\geq 0}\frac{c^m}{m!}K_a^m, \qquad K_a = I + Q_a/n. \tag{S40}$$

The failure state has a virtual self-loop. From vacuum, exactly one clock event followed by no others has probability ce^{-c}. Its pump probability is a, and its endpoint has $N = d$. Therefore, with $q_a = \Pr_a(N > 0)$,

$$ace^{-c} \leq q_a \leq a(1 - e^{-c}). \tag{S41}$$

The upper bound follows because the first actual event from vacuum must be a pump; otherwise the trial fails. At $a = 0$, $q_0 = 0$. For $c = 1$, declaring a nonzero pump upon observing any positive-number outcome gives zero false positives and miss probability at most $e^{-Ma/e}$.

One pump followed by j losses also contributes at first order. Define

$$w_j = e^{-c}\frac{c^{j+1}}{(j+1)!}\frac{d!}{(d-j)!n^j}, \qquad A = \sum_{j=0}^{d-1} w_j, \tag{S42}$$

where $0 \le j \le d$. For fixed n, d, c, the endpoint probabilities obey

$$\begin{aligned} p_a(d-j) &= aw_j + O(a^2), && j < d, \\ p_a(0) &= e^{-c} + aw_d + O(a^2), \\ p_a(f) &= 1 - e^{-c} - a(A + w_d) + O(a^2). \end{aligned} \tag{S43}$$

The coefficient can be written

$$A = \frac{ne^{-c}}{d+1}\left[(1+c/n)^{d+1} - 1 - (c/n)^{d+1}\right]. \tag{S44}$$

It satisfies $ce^{-c} \le A \le 1 - e^{-c}$ and approaches ce^{-c} at fixed d as $n \to \infty$. Consequently $H^2(p_0, p_a) = Aa/2 + O(a^2)$ and $F_a = A/a + O(1)$. These are fixed-time rare-event statistics. For pump amplitude $g = \sqrt{a}$, $F_g = 4aF_a \to 4A$, while relative-rate information $a^2F_a \to 0$.

For $N = k > 0$, the retained distribution has a nonnegative expansion $p_a(k) = \sum_{m\ge1} b_{k,m}a^m$. Thus $p_{a_1}(k) \ge (a_1/a_0)p_{a_0}(k)$ when $0 < a_0 < a_1 \le 1$. Keeping only these outcomes in the Hellinger sum gives

$$\begin{aligned} H^2(p_{a_0}, p_{a_1}) &\ge \frac{(\sqrt{a_1} - \sqrt{a_0})^2}{2}\frac{q_{a_0}}{a_0} \\ &\ge \frac{ce^{-c}}{2}(\sqrt{a_1} - \sqrt{a_0})^2. \end{aligned} \tag{S45}$$

The $a_0 = 0$ case follows directly from the positive-number outcomes. Taking $c = 1$ gives the attainable bound $H^2 \ge (\sqrt{a_1} - \sqrt{a_0})^2/(2e)$ for all n, d. Two strictly positive strengths with fixed ratio also approach the rate boundary: $H^2(p_a, p_{\lambda a}) = Aa(\sqrt{\lambda} - 1)^2/2 + O(a^2)$. A fixed interior base point is a different local limit.

VII. GEOMETRIC-CHANNEL BOUND AND MATCHING RESOURCE ORDER

The quantum output in this section is explicitly the retained system or a single erasure flag. No emitted field or additional rejected quantum state is available. Let $\mathcal{A}_a(\tau)$ be the retained map and write its dimensionless generator as $\mathcal{K}_0 + a\mathcal{J}$, where

$$\mathcal{J}(\rho) = \frac{n}{\binom{n}{d}}\sum_{|I|=d} c_I^\dagger \rho c_I. \tag{S46}$$

The semigroup of $\mathcal{K}_0$ and the map $\mathcal{J}$ are completely positive. Hence every coefficient in the Dyson expansion

$$\mathcal{A}_a(\tau) = \sum_{m\ge0} a^m \Phi_m(\tau) \tag{S47}$$

is CP. In finite dimension the expansion converges at finite time. Since $\mathcal{A}_1$ is trace nonincreasing, each partial sum $\Psi_j = \sum_{m=0}^{j}\Phi_m$ is also trace nonincreasing. Indeed, $\mathcal{A}_1 - \Psi_j$ is CP, so $0 \le \Psi_j^\dagger I \le \mathcal{A}_1^\dagger I \le I$.

Define the parameter-independent CPTP channel

$$\mathcal{D}_j(\rho) = \Psi_j(\rho) \oplus \mathrm{Tr}[(I - \Psi_j^\dagger I)\rho]|f\rangle\langle f|. \tag{S48}$$

Using $\sum_{j\ge m}(1-a)a^j = a^m$ gives, for $0 \le a < 1$,

$$\mathcal{E}_a = \mathcal{A}_a \oplus \text{erasure} = \sum_{j\ge0}(1-a)a^j\mathcal{D}_j. \tag{S49}$$

Thus the completed channel is a parameter-independent processing of a classical geometric program, $g_a(j) = (1-a)a^j$. This representation makes channel-simulation data processing [8] applicable to the specified endpoint output independently of an EP.

For a common input, including a retained ancilla, let $\omega_a = (\mathcal{E}_a \otimes \mathrm{id})(\rho)$. The root fidelity $f(\rho,\sigma) = \|\sqrt{\rho}\sqrt{\sigma}\|_1$ obeys

$$f(\omega_{a_0}, \omega_{a_1}) \ge G(a_0, a_1) = \frac{\sqrt{(1-a_0)(1-a_1)}}{1 - \sqrt{a_0 a_1}}. \tag{S50}$$

Every endpoint POVM has classical BC at least this large. Quantum Fisher information bounds the classical Fisher information of any POVM in the local estimation problem [9, 10]. Likewise,

$$F_Q[\omega_a] \le F[g_a] = \frac{1}{a(1-a)^2}, \qquad 0 < a < 1. \tag{S51}$$

These bounds are uniform in waiting time and finite system size. They constrain information per complete preparation, rather than per retained sample.

Writing $\Delta_{\sqrt{a}} = \sqrt{a_1} - \sqrt{a_0}$ gives

$$1 - G = \frac{\Delta_{\sqrt{a}}^2}{(1 - \sqrt{a_0 a_1})[1 - \sqrt{a_0 a_1} + \sqrt{(1-a_0)(1-a_1)}]}. \tag{S52}$$

Together with Eq. (S45),

$$\frac{\Delta_{\sqrt{a}}^2}{2e} \le \sup_{\rho_0,\tau}{}^{N/f} H^2 \le \frac{\Delta_{\sqrt{a}}^2}{2(1-a_1)^2}. \tag{S53}$$

The right bound also constrains arbitrary endpoint quantum readouts through Eq. (S50). For M independent preparations, root fidelities multiply; a final joint quantum measurement is permitted. Fidelity–trace-distance inequalities [11] then give a lower error bound using G^{2M}.

Vacuum number measurement supplies an attainable upper error bound. With fixed $a_1 \le a_{\max} < 1$ and fixed target error $0 < \varepsilon < 1/2$, these establish

$$M_{\rm opt} = \Theta(\Delta_{\sqrt{a}}^{-2}). \tag{S54}$$

The constants depend on the target error and $a_{\max}$, but not on n, d. They do not determine the exact optimal input or measurement.

For a specified classical endpoint distribution, the standard bounds are

$$\frac{1-\sqrt{1-e^{-2M\xi}}}{2} \le P_e^*(M) \le \frac{1}{2}e^{-M\xi}. \tag{S55}$$

Thus a necessary count is $\lceil -\log[4\varepsilon(1-\varepsilon)]/(2\xi)\rceil$, and a sufficient count is $\lceil -\log(2\varepsilon)/\xi\rceil$. These are bounds, not exact finite-sample optima. At $a_0 = 0$, the positive-number test supplies an alternative explicit upper bound.

The success-weighted postselection information bound and the information carried by outcome flags are established [12–14]. For classical N/f readout without an ancilla, the rejection branch contains only a flag, so

$$F_a = P_{\rm surv}F_{a|\rm surv} + \frac{(\partial_a P_{\rm surv})^2}{P_{\rm surv}(1-P_{\rm surv})}. \tag{S56}$$

For $n = 16, d = 3, a = .001, \tau = 1/16$, these contributions are 403.8220 and .70175, totaling 404.5238 per preparation. A divergence $F_a \sim 1/a$ does not provide infinite relative precision or zero-cost detection.

The vacuum protocol uses total evolution time $M/(n\gamma)$ and nM mode preparations. Its total vacuum pump rate is $an\gamma$. These costs cannot be interchanged without a physical resource convention. Reset overhead, detector backgrounds, timing resolution, and implementation of the d-body pump are not specified here.

The analytic-jump assumptions in Ref. [15] apply to a different parameterization and output class. Our pump jump is proportional to $\sqrt{a}$ and is not analytic in a at zero. Equation (S49) does not bound access to the environment's purification or extra rejected quantum information. It also does not solve model selection between different fractional spectral laws. Estimating a and inserting it into a known spectral formula is conditional model inference.

A. A boundary of number-only readout

For vacuum, permutation invariance gives $\omega_a = \bigoplus_k p_a(k)P_k/\binom{n}{k} \oplus p_a(f)|f\rangle\langle f|$. The block states are independent of a, making N/f sufficient for that input. This property fails for general preparations.

For $n = d = 2$, $H = 0$, $\rho_0 = |10\rangle\langle 10|$, and $z = 2a$, two outcomes absent at $a = 0$ satisfy

$$\begin{aligned} p_a(11) &= ae^{-2\tau}\tau^2 + O(a^2), \\ p_a(01) &= ae^{-2\tau}\tau^3/3 + O(a^2). \end{aligned} \tag{S57}$$

TABLE I. Endpoint discrimination at $d = 3$. The count $M_{\rm suff}$ ensures average error at most .05 through Eq. (S55). All optimized inputs in the table are vacuum.

n	(a_0, a_1)	$n\tau$	ξ	$M_{\rm suff}$
8	$(0, .001)$	1.2200	2.25571×10^{-4}	10208
16	$(0, .001)$	1.1021	2.02941×10^{-4}	11347
32	$(0, .001)$	1.0492	1.93083×10^{-4}	11926
8	$(.001, .002)$	1.2207	3.87572×10^{-5}	59411
16	$(.001, .002)$	1.1035	3.48951×10^{-5}	65986
32	$(.001, .002)$	1.0510	3.32143×10^{-5}	69326

Total number merges 01 with the already populated 10. Consequently,

$$\begin{aligned} H^2_{\rm modes} &= \tfrac{1}{2}ae^{-2\tau}(\tau^2 + \tau^3/3) + O(a^2), \\ H^2_N &= \tfrac{1}{2}ae^{-2\tau}\tau^2 + O(a^2). \end{aligned} \tag{S58}$$

Mode-resolved detection strictly improves the leading information for this fixed input. Therefore Eq. (S39) is not a general quantum-input optimization theorem, even though Eq. (S50) still bounds both readouts.

B. Numerical discrimination checks

The main scan covers $n = 8, 16, 32$, $d = 2, 3, 4$, and $a = 10^{-2}, 10^{-3}, 10^{-4}$ for 0 versus a and a versus $2a$. Each of the 54 cases enumerates $k = 0, \ldots, n$ and 65 logarithmic times on $.01/n \le \tau \le 30$, followed by local refinement for each k. A separate 257-point grid does not improve any refined value. All cases select vacuum with $1.0313 \le n\tau \le 1.3275$; the fixed protocol $\tau = 1/n$ attains at least 96.38% of the numerical maximum ξ. These searches are not global optimization certificates.

Nine additional searches use $d = 3$, $a_0 = .1$, and $a_1 = .1 + \delta$ at the same three sizes. As δ decreases from .01 to .0001, ξ/δ^2 approaches .6402, .6129, and .6053 for $n = 8, 16, 32$, respectively. This agrees with regular interior-point discrimination. Both-positive hypotheses approaching zero and a fixed positive base point are distinct limits.

For $n = 12, d = 3$, $z_0 = 1/144$ and $z_1 = 1.2z_0$, the optimized vacuum protocol has $\tau = .10652$ and $\xi = .000307870$. Equation (S55) gives a sufficient count of 7480. For the original filling $p = .75$ and the same N/f readout at $\tau = 5$, $\xi = 1.27995 \times 10^{-16}$. Its necessary and sufficient counts are approximately 6.49×10^{15} and 1.80×10^{16}, respectively. This compares the same parameter-discrimination task, not the accuracy of extracting a spectral exponent. The imbalance bias does not affect total-number or failure statistics.

VIII. VERIFICATION AND NUMERICAL BOUNDARIES

The reproducible bundle contains code, numerical tables, source records, and explicit checks. Exact symbolic characteristic polynomials were checked in six cases. The complete complex-fermion superoperator was compared with the frozen-set sign gauge for every $1 \le d \le n \le 4$, including random complex Hermitian h; maximum matrix residual was 1.96×10^{-14}. Additional checks include 4,134 integer CAR signs, spin intertwiners through seven modes, and 49 principal-submatrix identities.

The observable-selection checks test both the exact channel identities and their action on a physical state. The integer identities (S8) hold for all $2 \le n \le 7$, $1 \le d \le n$ (27 cases). Full positive-density-matrix propagation for $(n, d) = (4, 2), (4, 3), (5, 2)$ agrees with the population reduction retaining two marked modes, with maximum absolute error 1.34×10^{-15}; initial odd overlap and even–odd mixing matrix elements vanish. At $(4, 3)$ a direct double-precision eigensolver splits a residual zero Jordan block, producing a 5.53×10^{-6} error in the extracted even gap. This eigensolver comparison is unsuitable for gap validation; the exact matrix identities and direct time propagation remain valid. The other two direct even-gap comparisons agree within 3.2×10^{-14}.

For the three larger dynamical examples, both exact two-channel formulas agree with marked-mode propagation. Fits on $5 \le \gamma t \le 8$ check the predicted tail rates to within 0.23%; fitted parameters do not enter the figures' theoretical curves. The spectral-amplitude approximations on $6 \le \gamma t \le 9$ differ from the propagated signals by at most 0.21%. At very long times, subtracting the stationary value limits double-precision signal accuracy, so no claim is based on indefinitely extending these tails.

Perron roots for $d = 2, 3, 4$ at 11 sizes from $n = 16$ to 512, together with sizes $n-1, n-2$, use 99 converged logarithmic positive iterations. The maximum relative Collatz width is 2.00×10^{-12}. These are floating-point convergence diagnostics, not interval-arithmetic enclosures. Independent 50- and 70-digit spectra at $n = 16, 32, 48, 64$ give spacings agreeing within 10^{-10} after conversion to stored double precision. Agreement after rounding is not a certification of 50 accurate digits.

The endpoint precheck uses nonnegative uniformization with scaling and squaring. On this host, some long-time calls to a BLAS-based matrix exponential emitted floating-point warnings; those calls are not retained as validation evidence. The stable implementation agrees with independent short-time exponentials within 3.03×10^{-15}. Four full-CAR propagation tests, including a coherent even input, agree within 7.22×10^{-16}. Nine one-pump coefficient checks at $a = 10^{-7}$ differ relatively by at most 4.54×10^{-8}. Fisher derivatives from a Fréchet matrix exponential agree with finite differences within 1.69×10^{-9} relatively.

A separate $n = d = 2$ channel check constructs Dyson coefficients through order 12. Their Choi matrices and the completed partial-sum channels are positive to floating-point precision, with minimum eigenvalue -2.17×10^{-16}. Geometric-mixture residuals lie within the omitted-tail bound plus numerical tolerance. The mode-resolved counterexample is also verified directly. These finite checks corroborate the analytic construction; they do not replace its proof. All source data and code are included in the accompanying reproducibility bundle.

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